Spatiotemporal Vector and Magnitude Spectrum Visualization of Electromagnetic Fields for Specific Waveguide and Optical Fiber Modes

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Abstract

This program provides a three-dimensional computational tool for plotting electromagnetic fields in various guiding structures. The user can select the structural type (various waveguides or optical fibers) and specify geometric and environmental parameters, operating frequency, and the desired propagation mode (TE or TM). The simulator renders the fields both as vector plots and magnitude spectra across the cross-sectional dimensions (extending up to 2.5 times the radius for optical fibers) and along the propagation axis (spanning three wavelengths or periods). To enhance usability and facilitate intuitive field visualizations, a straightforward graphical user interface (GUI) has been integrated into the software.

Motivation

The primary motivation for this project stems from the objective to develop a simulation framework, analogous to conventional MATLAB-based simulators, capable of visualizing fundamental propagation modes in waveguides and optical fibers. This is achieved by defining the physical parameters of the medium and establishing appropriate initial conditions that induce the desired modes within the structure.

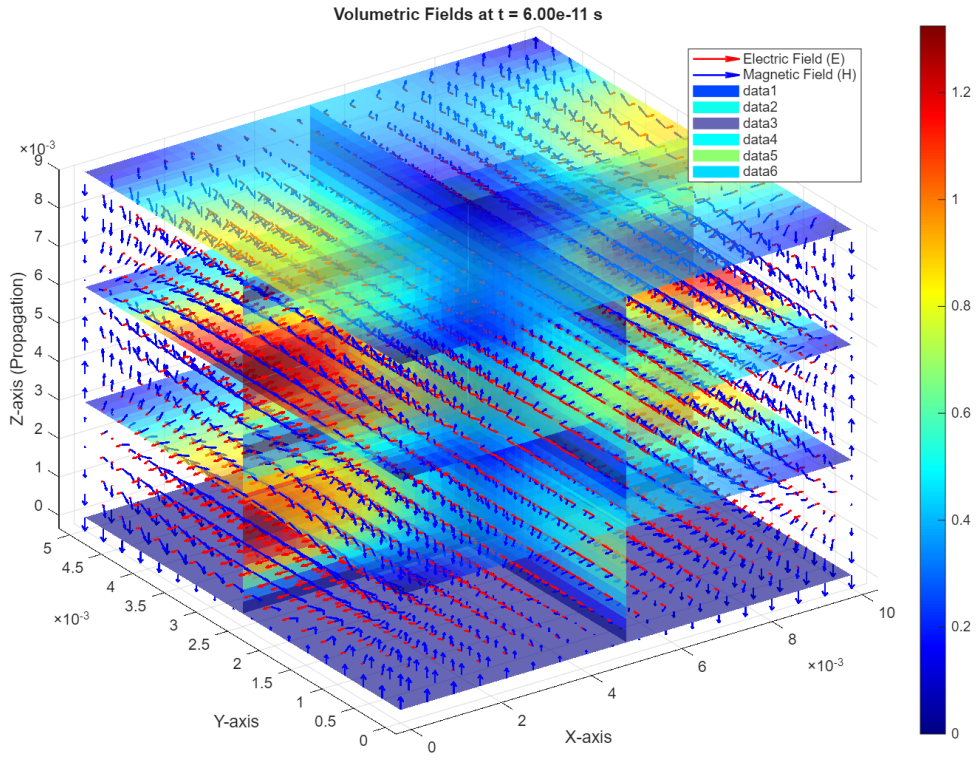


Figure 1: Volumetric field distribution for a rectangular waveguide ($a = 0.01$ m, $b = 0.005$ m) operating at 50 GHz in the TE_{11} mode, captured at $t = 3T$.

Program Features

Supported Modes

- **Parallel-plate, rectangular, and cylindrical waveguides:** Comprehensive support for all TE and TM modes.
- **Optical Fibers:** Supports TE_{0n} and TM_{0n} modes (restricted to separable propagating modes).

Visualization Capabilities

- 3D spatiotemporal plotting with a frame rate of 40 fps.
- Dense, localized spatial meshing to ensure high accuracy and comprehensive field coverage within the medium.
- High execution speed achieved through optimized numerical field computation.

User Interface (UI)

- Persian-language menu system for configuring structural types, physical and geometric parameters, operating frequency, and propagation modes.
- Robust error handling to manage non-existent modes or non-propagating states based on the specified cutoff frequencies.

- Navigation capability allowing users to return to previous menus for input correction.
- The software provides a user-friendly and intuitive experience, serving as an effective educational tool for exploring electromagnetic fields in waveguides and fibers.

Mode Examples in Various Structures

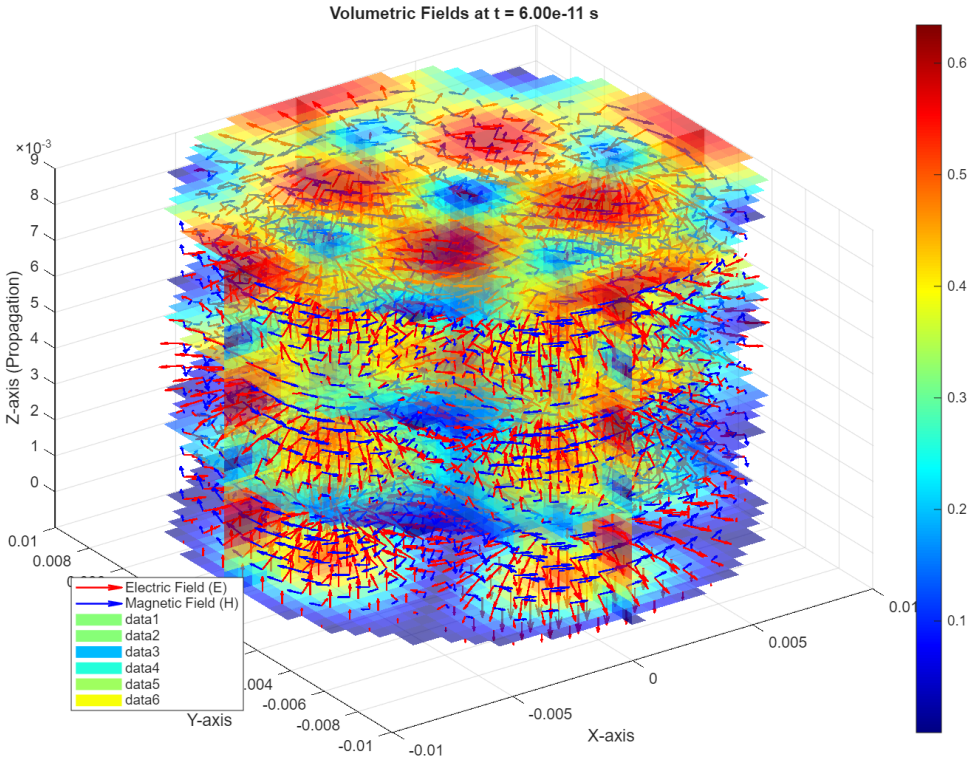


Figure 2: TM_{21} mode in a circular waveguide with a radius of 0.01 m operating at 50 GHz.

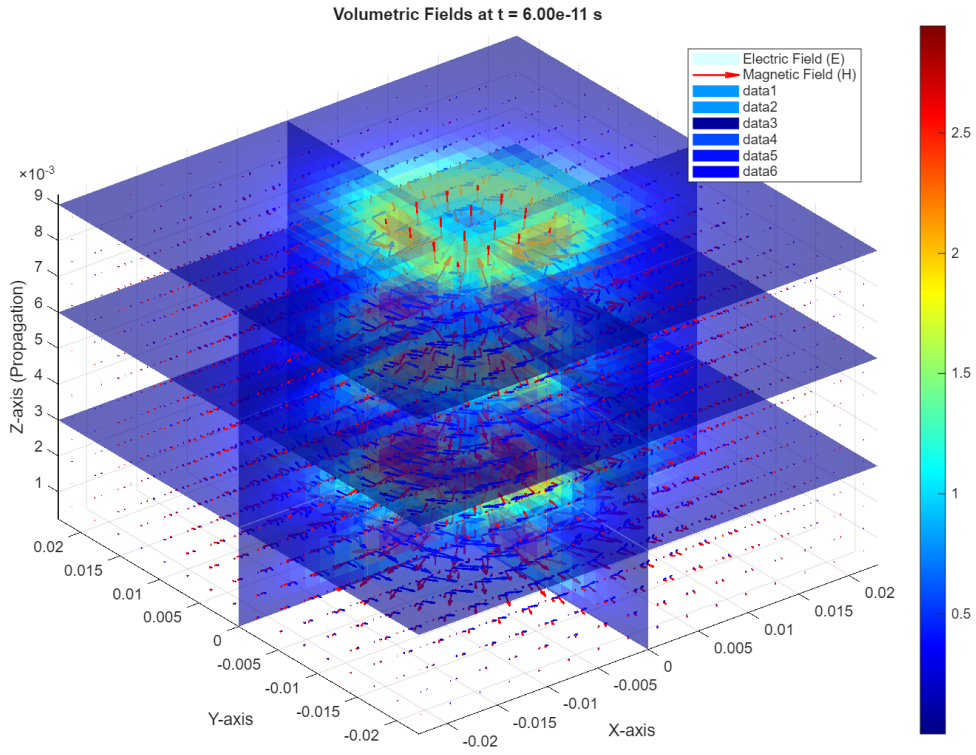


Figure 3: TM_{01} mode in an optical fiber (core radius = 0.01 m, core relative permittivity $\epsilon_{r1} = 2$, cladding relative permittivity $\epsilon_{r2} = 1.9$) at 50 GHz.

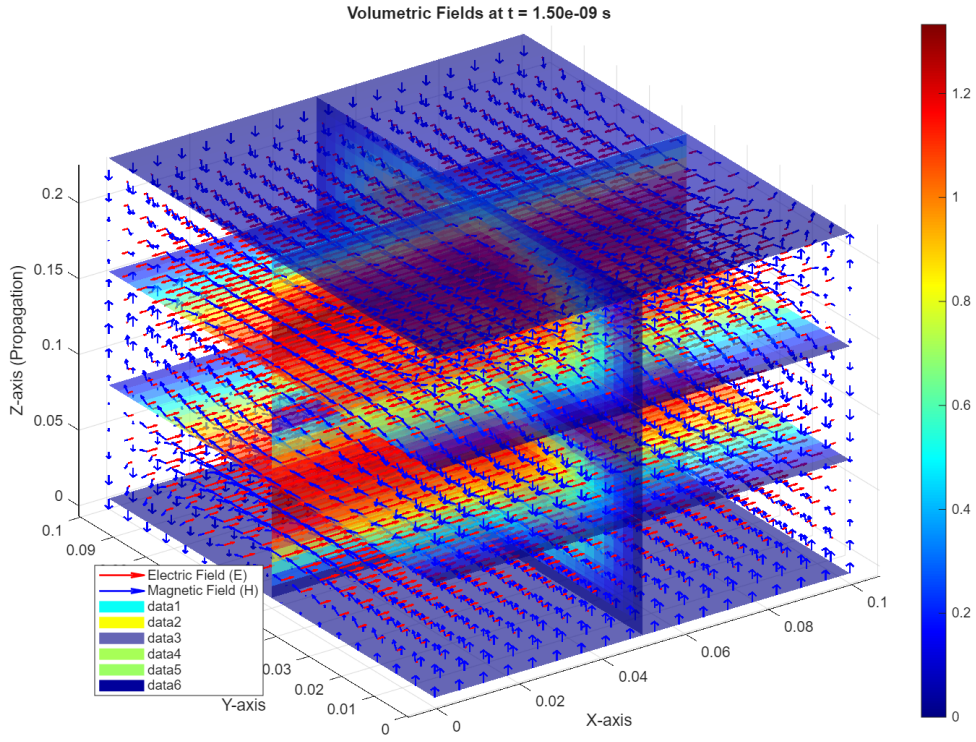


Figure 4: TE_1 mode in a parallel-plate waveguide with a plate separation of $b = 0.1$ m operating at 2 GHz.